

A 13-month-old girl is brought to the physician because of a pruritic rash for 2 days. The girl's mother says she noticed a few isolated skin lesions on her trunk two days ago that appear to be itching. The girl received her routine immunizations 18 days ago. Her mother has been giving her ibuprofen for her symptoms. The patient has no known sick contacts. She is at the 71st percentile for height and the 64th percentile for weight. She is in no acute distress. Her temperature is 38.1°C (100.6°F), pulse is 120/min, and respirations are 26/min. Examination shows a few maculopapular and pustular lesions distributed over the face and trunk. There are some excoriation marks and crusted lesions as well. Which of the following is the most likely explanation for these findings?

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✍️ 💬 ↶

- ☐ A Immune complex formation and deposition
- ☐ B Antigen contact with presensitized T-lymphocytes
- ☐ C IgM and IgG antibodies binding to cellular antigens
- ☐ D Reactivation of virus dormant in dorsal root ganglion
- ☐ E Crosslinking of preformed IgE antibodies
- ☐ F Replication of the attenuated vaccine strain

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The simultaneous appearance of various stages of lesions (papules, vesicles, and crusts) is the hallmark of one of the diseases (in its fully developed form) for which this girl was immunized.

**13% A** Immune complex formation and deposition

Immune complex formation and deposition describes the pathophysiology of type III hypersensitivity reactions. An example of this reaction and a rare complication of tetanus and diphtheria vaccinations is the Arthus reaction, which can lead to a localized reaction involving pain, edema, hemorrhage, and sometimes necrosis 4-12 hours after vaccination. It would be very unusual for the Arthus reaction to produce a disseminated rash 16-18 days after vaccination.

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15% B Antigen contact with presensitized T-lymphocytes

Antigen contact with presensitized T-lymphocytes is the mechanism underlying type IV hypersensitivity reactions. These hypersensitivity responses classically appear 48-72 hours after exposure to the antigen. Clinical presentation is typically erythematous, indurated tissue at the site of exposure. Contact dermatitis (e.g., due to poison ivy or nickel) is a well known example of a type IV hypersensitivity reaction. This patient's disseminated rash following a vaccination is not consistent with this immunological mechanism.

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9% C IgM and IgG antibodies binding to cellular antigens

IgM and IgG antibodies binding to cellular antigens is responsible for type II hypersensitivity reactions. Pertinent examples of



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9% **C** IgM and IgG antibodies binding to cellular antigens

IgM and IgG antibodies binding to cellular antigens is responsible for type II hypersensitivity reactions. Pertinent examples of such reactions are ABO incompatibility and Myasthenia gravis. Reports of type II hypersensitivity reactions accounting for vaccine-related adverse events are very rare. When they do occur, a localized eczematous rash at the injection site several hours after injection is typical.

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8% **D** Reactivation of virus dormant in dorsal root ganglion

Infection following reactivation of the varicella zoster virus (chicken pox) dormant in dorsal root ganglion is the mechanism by which shingles arises. A shingles rash classically presents in a dermatomal distribution (typically of 1–3 dermatomes) on one side of the body with severe pain, which is not consistent with this patient's rash. Moreover, she has no history of infection with varicella zoster virus, which rules out shingles.



» Feedback

7% **E** Crosslinking of preformed IgE antibodies

Crosslinking of preformed IgE antibodies is the mechanism underlying type I hypersensitivity reactions. Type I hypersensitivity reactions manifest as anaphylaxis. This patient is instead presenting with pruritus, a maculopapular rash, and crusted lesions.

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48% **F** Replication of the attenuated vaccine strain

Replication of an attenuated viral strain in a vaccine is the most likely explanation for this patient's presentations. The patient likely received a varicella vaccine (other common vaccines given at 12 months include Hib, MMR, and polio) and is showing an attenuated form of chickenpox with pruritic rash, maculopapular lesions, and crusting. Roughly 3% of patients will demonstrate a mild varicella-like rash after the initial vaccine dose but they will not be contagious. A varicella-like rash can be due to either wild-type or vaccine-type virus. A rash that occurs more than 2 weeks after varicella vaccination in patients with no known exposure to infected individuals is most likely caused by replication of the vaccine-type virus strain.

8% **D** Reactivation of virus dormant in dorsal root ganglion



Shingles of the upper extremity

Left arm of a child showing multiple, uniform vesicles on an erythematous base. The dermatomal distribution (C8 and T1) is characteristic of shingles. Patients complain of severe pain prior to the appearance of lesions.



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A 45-year-old man comes to the physician for the evaluation of painful swallowing and retrosternal pain over the past 2 days. He was recently diagnosed with HIV infection, for which he now takes tenofovir, emtricitabine, and raltegravir. There is no family history of serious illness. He has smoked one pack of cigarettes daily for the past 20 years. He drinks 2-3 beers per day. He does not use illicit drugs. Vital signs are within normal limits. Physical examination of the oral cavity shows no abnormalities. The patient's CD4+ T-lymphocyte count is $80/\text{mm}^3$ (normal ≥ 500). Empiric treatment is started. Two weeks later, he reports no improvement in his symptoms. Esophagogastroduodenoscopy is performed and shows multiple well-circumscribed, round, superficial ulcers in the upper esophagus. Which of the following is the most likely underlying cause of this patient's symptoms?



» Feedback



- ☐ A Infection with herpes simplex virus
- ☐ B Adverse effect of medication
- ☐ C Transient lower esophageal sphincter relaxation
- ☐ D Allergic inflammation of the esophagus
- ☐ E Degeneration of inhibitory neurons within the myenteric plexuses
- ☐ F Diffuse esophageal spasm
- ☐ G Infection with Candida species
- ☐ H Infection with cytomegalovirus



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A 45-year-old man comes to the physician for the evaluation of **painful swallowing** and retrosternal pain over the past 2 days. He was recently diagnosed with **HIV infection**, for which he now takes tenofovir, emtricitabine, and raltegravir. There is no family history of serious illness. He has smoked one pack of cigarettes daily for the past 20 years. He drinks 2-3 beers per day. He does not use illicit drugs. Vital signs are within normal limits. Physical examination of the oral cavity shows no abnormalities. The patient's CD4+ T-lymphocyte count is $80/\text{mm}^3$ (normal ≥ 500). **Empiric treatment is started.** Two weeks later, he reports no improvement in his symptoms. Esophagogastroduodenoscopy is performed and shows multiple **well-circumscribed, round, superficial ulcers in the upper esophagus**. Which of the following is the most likely underlying cause of this patient's symptoms?

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This patient is immunosuppressed and presents with odynophagia, which suggests esophagitis. Empiric treatment of esophagitis in HIV-positive patients typically consists of fluconazole, which was already started in this patient.



45% **A** Infection with herpes simplex virus ✓

Herpes simplex virus (HSV) esophagitis can occur in patients with HIV and CD4 counts $< 100/\text{mm}^3$ and is considered an AIDS-defining condition. Typical EGD findings include multiple round, superficial ulcers and friable mucosa. Biopsy with viral culture or PCR can confirm the diagnosis, and treatment consists of acyclovir. In this case, the patient was appropriately treated with empiric fluconazole (esophageal candidiasis is much more common than HSV esophagitis). However, the lack of clinical response prompted further investigation.



HIV-associated conditions

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5% **B** Adverse effect of medication

Drug-induced esophagitis (also referred to as pill esophagitis) is caused by prolonged contact with caustic medications, leading to esophageal mucosal irritation, erosion, and ulceration. Commonly implicated medications include antibiotics (tetracycline, doxycycline, clindamycin), NSAIDs, bisphosphonates, and iron pills. None of this patient's current medications are known to cause esophagitis. EGD findings in a patient with pill esophagitis typically demonstrate punched-out mucosa with inflammation of the surrounding mucosa.

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1% **C** Transient lower esophageal sphincter relaxation

Gastroesophageal reflux disease (GERD) is caused by transient lower esophageal sphincter relaxation and reflux of stomach contents into the esophagus. Symptoms are usually chronic and episodic in nature and can include heartburn, regurgitation sensation, chronic cough, and retrosternal chest pain. Dysphagia and odynophagia are not typical and are considered red flag symptoms which should prompt further evaluation (usually with EGD). This patient's EGD findings are not typical for GERD and his HIV-positive status makes a different etiology more likely.

> Feedback

1% **D** Allergic inflammation of the esophagus

Eosinophilic esophagitis is a poorly understood type of esophagitis that likely occurs in response to allergens and is mediated by eosinophil release. Diagnosis consists of identification of mucosal eosinophilia on esophageal biopsies. Most patients present with recurrent dysphagia and food impaction, rather than odynophagia. Typical findings on EGD include circumferential mucosal lesions and mucosal fragility.



> Feedback

1% **E** Degeneration of inhibitory neurons within the myenteric plexuses

Achalasia, caused by the degeneration of inhibitory neurons in the myenteric plexus of the esophagus, may present with dysphagia, retrosternal pain, and regurgitation. EGD findings are typically normal, and the diagnosis is confirmed with manometry or barium esophagram (classic "bird-beak" appearance). This patient's odynophagia and presence of ulcers on EGD suggest a different diagnosis.

> Feedback

1% **F** Diffuse esophageal spasm

Diffuse esophageal spasm (DES) may present with dysphagia and squeezing retrosternal chest pain, though there are often predictable triggers (e.g., caffeine, hot or cold foods). EGD would show normal esophageal mucosa and the diagnosis is confirmed with barium esophagram (classic finding is corkscrew esophagus). This patient's odynophagia and ulcers on endoscopy are not consistent with DES and suggest a different diagnosis.



Caption



Eosinophilic esophagitis

Endoscopy of the esophagus showing circumferential mucosal lesions (rings/corrugations), a typical finding in eosinophilic esophagitis.

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13% G Infection with Candida species

Candida species cause the vast majority of HIV-associated esophagitis, and for this reason, patients with HIV and odynophagia are usually treated empirically with fluconazole. This patient's lack of clinical response to empiric fluconazole makes esophageal candidiasis unlikely. Additionally, EGD findings in esophageal candidiasis typically include whitish/yellow plaques lining the esophageal mucosa, unlike the superficial ulcers seen in this patient. Lastly, oral thrush is usually also present.

[» Feedback](#)

33% H Infection with cytomegalovirus

Cytomegalovirus (CMV) infection is an important cause of esophagitis to consider in immunocompromised patients. However, CMV infections typically only occur when CD4 counts are $< 50/\text{mm}^3$. Classic EGD findings are deep linear or longitudinal ulcers, unlike the round, superficial ulcers seen in this patient.

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Candida esophagitis

Endoscopic image of the esophagus showing streaky, confluent, white-grayish lesions, which consist of fungi, fibrin, and flaked off squamous cells. Lesions are difficult to scrape off and are typical of Candida esophagitis.

Our great thanks to Dr. Merkel, Dr. Wiens, Dr. Kämmer (Center for Internal Medicine, St. Hedwig Hospital, Berlin) for kindly providing this image.





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An 18-year-old man is brought to the emergency department by his girlfriend because of a pruritic rash on his penis that has been present for the past hour. The rash developed shortly after the patient had protected sexual intercourse with his girlfriend for the first time. His girlfriend does not have any symptoms. Five days ago, the patient visited a friend who was complaining of intense pruritus on her elbows, wrists, groin, and axillae. The friend subsequently underwent treatment that required her to wash her bedding, clothing, and towels in hot water. Two years ago, the patient developed an itchy rash around his mouth after blowing up balloons at a birthday party. He is breathing comfortably. Vitals signs are within normal limits. Examination shows multiple well-circumscribed, raised, erythematous plaques with mild excoriations that extend from the base to the glans of the penis. The remainder of the examination shows no abnormalities. Administration of which of the following is the most appropriate next step in management?



» Feedback



- ☐ A | Intravenous acyclovir
- ☐ B | Topical permethrin
- ☐ C | Oral cromolyn sodium
- ☐ D | Subcutaneous epinephrine
- ☐ E | Oral famotidine
- ☐ F | Intravenous prednisone
- ☒ G | Oral cetirizine



Show answer

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[» Feedback](#)

The eruption of an urticarial rash on the patient's penis shortly after protected sexual intercourse and history of a rash following exposure to balloons suggest a hypersensitivity reaction to latex.

**4% A Intravenous acyclovir**

Intravenous acyclovir is used to treat certain viral infections, e.g., herpes simplex virus infection. While HSV infection should be considered as a differential diagnosis in a sexually active patient presenting with a genital rash, the timing and appearance of his rash are not consistent with this diagnosis (HSV typically appears 2-7 days after exposure and is characterized by vesicles and ulcerations). Furthermore, oral acyclovir is preferred over IV administration for genital HSV infection.

[» Feedback](#)**22% B Topical permethrin**

Topical permethrin is used for the treatment of lice and scabies. This patient's friend has symptoms consistent with scabies, and while scabies is spread through close contact, the typical rash is characterized by elongated papules, red burrows, and scattered vesicles. This patient's rash is consistent with urticaria, against which permethrin is not effective.

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Caption



Skin lesion in scabies

Localization: periumbilical region

Lesion: papules, erythema, excoriations

Description: elongated, erythematous papules, inflamed nodules, and excoriations

Subcutaneous epinephrine



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Subcutaneous epinephrine is used for the treatment of anaphylaxis, a severe form of type 1 hypersensitivity reaction. While urticaria is a common feature in anaphylaxis, this patient has no other symptoms to suggest it (e.g., bronchospasm, laryngeal edema, abdominal pain, hypotension). A different therapy would be more appropriate for a mild to moderate localized reaction.

[Feedback](#)9% **E** Oral famotidine

Famotidine is an H₂ receptor blocker commonly used for the treatment of GERD and can also be used in the treatment of anaphylaxis. While acute-onset urticaria after exposure to a potential allergen should raise concern for anaphylaxis, this patient has no other symptoms to suggest anaphylaxis (e.g., bronchospasm, laryngeal edema, abdominal pain, hypotension) or GERD. A different therapy would be more appropriate.

[Feedback](#)10% **F** Intravenous prednisone

Prednisone is a potent anti-inflammatory that has many uses, including the treatment of certain hypersensitivity reactions. Because glucocorticoids can cause serious systemic side effects, they are typically reserved for moderate to severe type 1 hypersensitivity reactions (e.g., with angioedema or pronounced urticaria). If glucocorticoids are used for treatment of urticaria, oral rather than intravenous administration is preferred. This patient has a mild to moderate localized hypersensitivity reaction, making another treatment more appropriate.

[Feedback](#)44% **G** Oral cetirizine

The patient's localized urticaria is consistent with a mild to moderate type 1 hypersensitivity reaction to latex, which means that a second-generation antihistamine such as cetirizine is indicated. Type 1 hypersensitivity reactions are caused by preformed IgE antibodies (from prior sensitization), which then activate rapid histamine release from mast cells upon re-exposure to the antigen. Cetirizine competes with histamine at the H₁ receptor to stabilize mast cells and is less sedating than first-generation antihistamines (e.g., diphenhydramine, chlorpheniramine, hydroxyzine). This patient should also be advised to avoid contact with latex in the future.



urticaria is a condition characterized by red, raised skin lesions (hives) and swelling (edema), often caused by an allergic reaction.

5% **E** Oral famotidine

Famotidine is a histamine H₂ receptor antagonist used to treat acid-related disorders such as gastroesophageal reflux disease (GERD) and peptic ulcers. It is not typically used for anaphylaxis. The patient has no other contraindications. A different treatment is needed for the acute allergic reaction.

10% **F** Intravenous prednisone

Prednisone is a corticosteroid used to reduce inflammation and suppress the immune system. It is often used for severe allergic reactions. Because glucocorticoids can suppress the immune system, they should be used cautiously in patients with certain infections. The patient has no other contraindications. Making another decision is needed.



44% **G** Oral cetirizine

The patient's history indicates that a second exposure to the allergen (latex) has occurred, leading to a preformed IgE-mediated allergic reaction. Re-exposure to the allergen can trigger a more severe reaction than first-generation antihistamines (like cetirizine) can manage. The patient should be advised to avoid further exposure to latex.



gest it (e.g., bronchospasm, laryngeal edema) or a mild to moderate localized reaction.

> Feedback

It can also be used in the treatment of allergic rhinitis. If there is a high concern for anaphylaxis, this patient should be monitored for signs of severe allergic reaction (e.g., abdominal pain, hypotension) or GERD.

> Feedback

For certain hypersensitivity reactions, corticosteroids are used for moderate to severe type 1 hypersensitivity reactions. Corticosteroids are used for treatment of urticaria, allergic rhinitis, and severe localized hypersensitivity reactions.

> Feedback

ivity reaction to latex, which means that severe allergic reactions are caused by the release of histamine from mast cells upon exposure to latex. Antihistamines (like cetirizine) can help stabilize mast cells and are less sedating than first-generation antihistamines (like diphenhydramine). This patient should also be advised to avoid further exposure to latex.

Acute urticaria

Severely pruritic, erythematous, slightly raised skin alterations due to upper dermal edema. The lesions are acute in onset and usually transient.

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Hypersensitivity reactions



Urticaria multiforme

Right palm of an 80-year-old patient: Multiple, partially confluent and partially solitary, pale reddish-brown, serpiginous lesions. The lesions were pruritic and disappeared after 12 hours. All findings indicate urticaria multiforme as a manifestation of a type I hypersensitivity reaction. In this case, the cause of the allergic reaction could not be identified.



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A 10-month-old boy is brought to the physician by his mother because of a 2-day history of rhinorrhea, nasal congestion, and cough. He has been feeding normally and has not had vomiting or diarrhea. The infant was born at term via uncomplicated spontaneous vaginal delivery. Immunizations are up-to-date. Eight months ago, he was treated for a urinary tract infection. Four months ago, he had an uncomplicated upper respiratory infection. He is alert and well-appearing. His temperature is 38.4°C (101.1°F), pulse is 110/min, respirations are 32/min, and blood pressure is 90/56 mm Hg. Examination shows erythematous nasal mucosa. Scattered expiratory wheezing is heard throughout both lung fields. The remainder of the examination shows no abnormalities. An x-ray of the chest is shown. After administration of an antipyretic, which of the following is the most appropriate next step in management?



» Feedback



- ☐ A | Provide reassurance
- ☐ B | Begin oral antibiotic therapy
- ☐ C | Perform PPD skin testing
- ☐ D | Obtain a thoracic CT scan
- ☐ E | Measure serum immunoglobulin levels
- ☐ F | Measure T cell count
- ☐ G | Perform bronchoscopy



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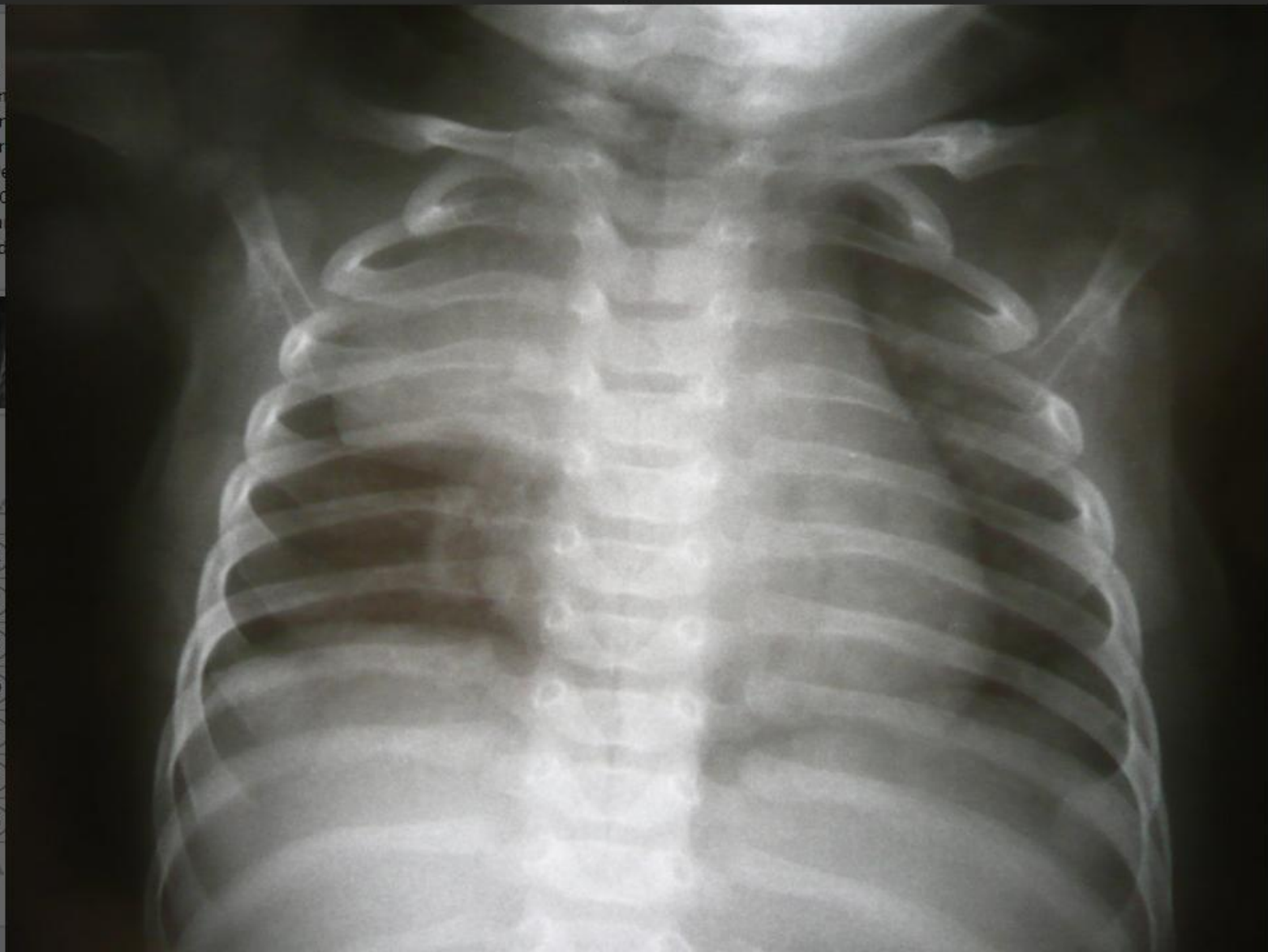
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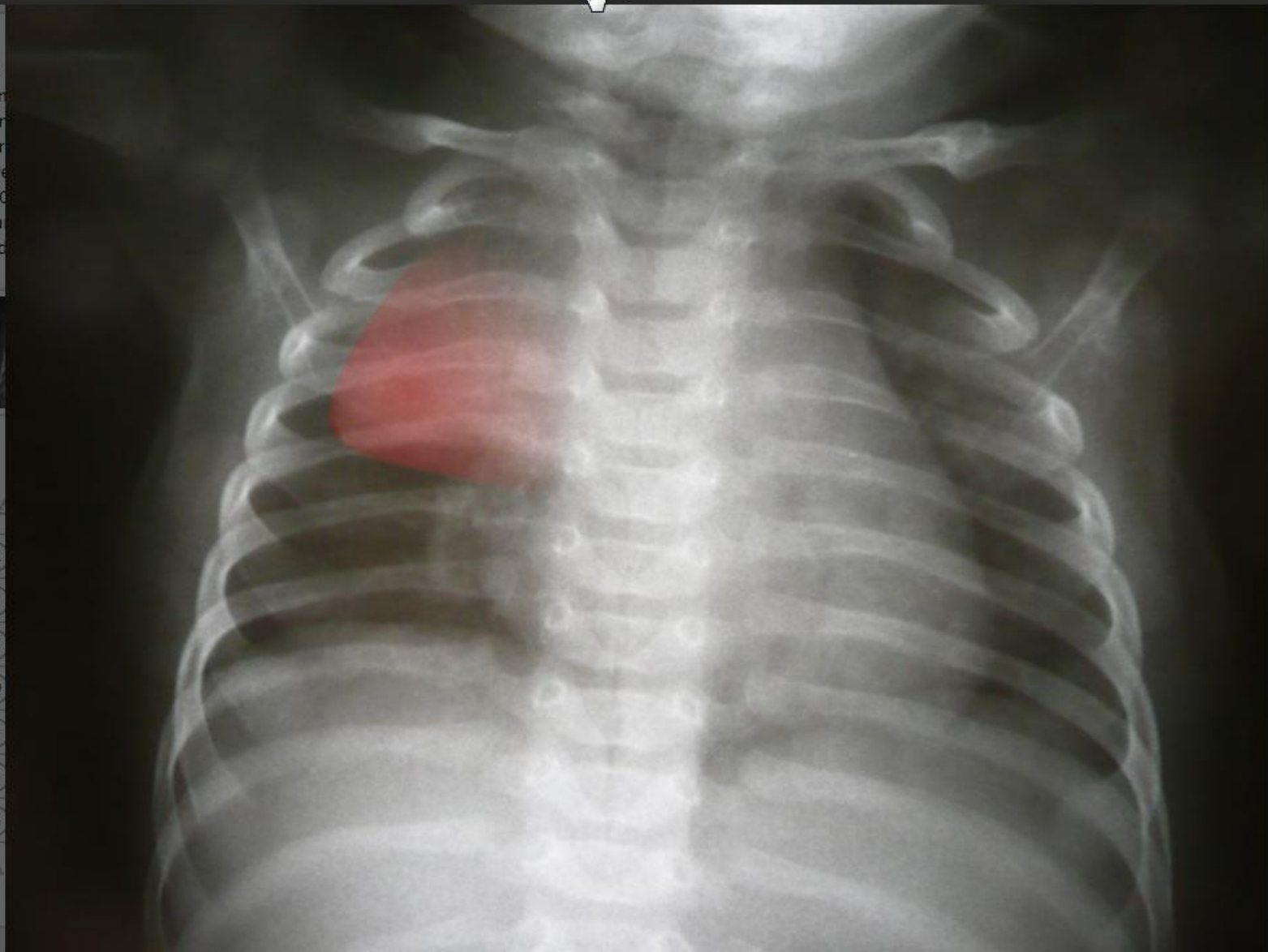
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This boy's presentation is suggestive of bronchiolitis. His chest x-ray shows an opacity in the right upper lung field, a common finding in children his age.



37% **A** Provide reassurance



Supportive therapy and reassurance are indicated for this boy's uncomplicated bronchiolitis. His vital signs are reassuring and his chest x-ray is consistent with thymic shadow, a normal finding in children less than 3 years old. The thymus varies in size and shape and involutes with age, eventually being replaced by fatty tissue.



Adaptive immune system

[» Feedback](#)

24% **B** Begin oral antibiotic therapy

2% **C** Perform PPD skin testing

A PPD skin test is a screening tool used to diagnose latent *Mycobacterium tuberculosis* (TB) infection, which is typically



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37%

A

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Adaptive immune system

[» Feedback](#)

24%

B

Begin oral antibiotic therapy

Oral antibiotics would be appropriate if bacterial pneumonia were suspected. Although this infant has fever and cough, the results of his chest x-ray are consistent with a benign finding that is typical of children his age. Apart from symptoms of viral bronchiolitis, he is otherwise healthy; antibiotics are not indicated at this time.

[» Feedback](#)

2%

C

Perform PPD skin testing

A PPD skin test is a screening tool used to diagnose latent *Mycobacterium tuberculosis* (TB) infection, which is typically asymptomatic. This patient has no reported risk factors for TB (e.g., close contact with an infected person, immigration from an endemic area) and screening is not indicated. He has bronchiolitis and the opacity seen on his x-ray is consistent with a benign finding that is common for his age.

[» Feedback](#)

4%

D

Obtain a thoracic CT scan

CT scan of the chest would be appropriate if there were pulmonary or mediastinal findings on chest x-ray that required further characterization (e.g., infiltrate, mediastinal mass). However, the opacity seen on this patient's x-ray is a benign finding that is common for his age. A CT scan carries a large radiation burden and there is no indication to warrant further imaging.

[» Feedback](#)

20%

E

Measure serum immunoglobulin levels

Measurement of serum immunoglobulin levels would be indicated if an immunodeficiency disorder or thymoma were suspected. There is no history to suggest thymoma or immunodeficiency in this patient; UTI and uncomplicated upper respiratory infection are common in healthy children. This infant's normal thymic shadow on chest x-ray is inconsistent with SCID and DiGeorge syndrome, which are typically characterized by thymic hypoplasia.

4. PPD skin test is a screening tool used to diagnose latent *Mycobacterium tuberculosis* (TB) infection, which is typically asymptomatic. This patient has no reported risk factors for TB (e.g., close contact with an infected person, immigration from an endemic area) and screening is not indicated. He has bronchiolitis and the opacity seen on his x-ray is consistent with a benign finding that is common for his age.

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10% **F** Measure T cell count

Measurement of T cell count would be indicated if an immunodeficiency disorder were suspected, such as SCID or DiGeorge syndrome. However, there is no history to suggest immunocompromise in this patient; UTI and uncomplicated upper respiratory infection are common in healthy children. The presence of a normal thymic shadow on this patient's chest x-ray is inconsistent with SCID and DiGeorge syndrome, which are characterized by thymic hypoplasia.

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3% **G** Perform bronchoscopy

Bronchoscopy is an invasive diagnostic tool that can be used to diagnose various bronchopulmonary diseases (e.g., suspected malignancy, atypical infection, tracheomalacia, diffuse alveolar hemorrhage). This patient has a fever and opacity on chest x-ray, but is an otherwise healthy child with a short duration of mild symptoms and reassuring vital signs. His radiographic opacity is consistent with a benign finding that is typical of children his age. Bronchoscopy is not indicated.

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Expanded answers

A 22-year-old man comes to the emergency department because of sudden onset of facial swelling, abdominal pain, and nausea 1 hour after eating some trail mix. He does not have shortness of breath or hoarseness. He has never had a similar episode before. He has asthma treated with an albuterol inhaler, which he uses less than once a month. His pulse is 90/min, respirations are 25/min, and blood pressure is 125/80 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. Physical examination shows perioral swelling and scattered urticaria over the face and the fingers of his right hand. There is no swelling of the oropharynx and no audible stridor. The lungs are clear to auscultation bilaterally. Treatment with intravenous diphenhydramine is begun. Shortly after, the abdominal pain and nausea have resolved, the swelling has improved, and the rash has started to fade. Which of the following is the most appropriate next step in management?

[» Feedback](#)

- ☐ (A) Recommend avoidance of peanuts
- ☐ (B) Perform patch test
- ☐ (C) Prescribe epinephrine autoinjector
- ☐ (D) Increase dosage of inhaled albuterol
- ☐ (E) Advice to carry oral cetirizine at all times
- ☐ (F) Administer oral prednisone

 [Show answer](#)

✖

A 22-year-old man comes to the emergency department because of sudden onset of facial swelling, abdominal pain, and nausea 1 hour after eating some trail mix. He does not have shortness of breath or hoarseness. He has never had a similar episode before. He has asthma treated with an albuterol inhaler, which he uses less than once a month. His pulse is 90/min, respirations are 25/min, and blood pressure is 125/80 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. Physical examination shows perioral swelling and scattered urticaria over the face and the fingers of his right hand. There is no swelling of the oropharynx and no audible stridor. The lungs are clear to auscultation bilaterally. Treatment with intravenous diphenhydramine is begun. Shortly after, the abdominal pain and nausea have resolved, the swelling has improved, and the rash has started to fade. Which of the following is the most appropriate next step in management?

[» Feedback](#)

This patient exhibits urticaria and swelling consistent with a food-induced type I hypersensitivity reaction, likely attributable to an ingredient of the trail mix.



20% **A** Recommend avoidance of peanuts

Although it might be appropriate for the patient to avoid peanuts, this recommendation is not sufficient given the possibility of a life-threatening reaction to subsequent exposure to the allergen. In addition, the exact nature of the allergen is unknown: peanuts could be the culprit, but another ingredient in the trail mix could have caused the type I hypersensitivity reaction.

[» Feedback](#)

24% **B** Perform patch test

The patch test is the gold standard for diagnosis of contact dermatitis, a type 4 hypersensitivity reaction. In a patch test, small amounts of potential contact allergens are applied to a patient's back for 48–72 hours to induce a local type 4 hypersensitivity reaction and identify the irritant. This patient's presentation is more consistent with a type 1 hypersensitivity reaction (not a type 4 hypersensitivity reaction), so a patch test would not be appropriate.

[» Feedback](#)

46% **C** Prescribe epinephrine autoinjector



Epinephrine is the primary treatment for anaphylaxis, and a patient with a history of a type I hypersensitivity reaction should be



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The patch test is the gold standard for diagnosis of contact dermatitis, a type 4 hypersensitivity reaction. In a patch test, small amounts of potential contact allergens are applied to a patient's back for 48-72 hours to induce a local type 4 hypersensitivity reaction and identify the irritant. This patient's presentation is more consistent with a type 1 hypersensitivity reaction (not a type 4 hypersensitivity reaction), so a patch test would not be appropriate.

[» Feedback](#)46% **C****Prescribe epinephrine autoinjector**

Epinephrine is the primary treatment for anaphylaxis, and a patient with a history of a type I hypersensitivity reaction should be prescribed an epinephrine autoinjector in case of repeat exposure to the allergen. Although the patient's reaction was relatively mild, a subsequent exposure could result in a more systemic reaction or life-threatening anaphylaxis as the severity of previous allergic reactions does not predict the severity of future reactions. The patient should also be referred to an allergist.

**Hypersensitivity reactions**[» Feedback](#)0% **D****Increase dosage of inhaled albuterol**

Despite a history of asthma, this patient has no signs or symptoms of bronchospasm associated with his presentation, so there is no need to change his asthma medication.

[» Feedback](#)4% **E****Advice to carry oral cetirizine at all times**

Although the patient responded well to an antihistamine, cetirizine's onset of action is around 20 minutes, which would be too long in the case of a more severe reaction in the future.

[» Feedback](#)6% **F****Administer oral prednisone**

Steroids, such as oral prednisone, may be used as adjunctive treatment for patients who do not respond completely to an antihistamine. This patient responds well to intravenous diphenhydramine and does not need steroids.

[» Feedback](#)

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[Expand answers](#)

A 27-year-old man comes to the emergency department for progressive weakness and numbness in his legs for 5 days. Symptoms originally started with tingling in both feet and have progressed to involve the knees and hips; he is currently unable to walk without assistance. Two weeks ago, the patient had diarrhea that subsided without antibiotics. Neurologic examination shows weakness, decreased sensation, and absent patellar reflexes in both lower extremities. Lumbar puncture shows elevated CSF protein and no white blood cells. Which of the following structures is most likely affected in this patient?



» Feedback



- ☒ A | Postsynaptic acetylcholine receptors
- ☐ B | Muscle endomysium
- ☐ C | Schwann cells
- ☐ D | Presynaptic terminals of cholinergic neurons
- ☐ E | Cerebral white matter



Show answer

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[» Feedback](#)

This patient with a history of recent diarrheal illness now presents with progressive ascending weakness. His exam findings and elevated CSF protein support a diagnosis of Guillain-Barré syndrome.



8% **A** Postsynaptic acetylcholine receptors

Autoantibodies produced against postsynaptic AChR are the underlying cause of myasthenia gravis. The binding of antibodies to these receptors causes competitive inhibition that decreases the excitatory action of ACh. It also results in ACh desensitization and can lead to the destruction of those receptors. As in this patient, MG can cause muscle weakness; however, patients with MG typically presents with ptosis, diplopia, and weakness that worsens with use (symptoms are typically the worse at the end of the day).

[» Feedback](#)

5% **B** Muscle endomysium

Inflammation of the endomysium is seen in patients with polymyositis. As a subtype of inflammatory myopathies, polymyositis is characterized by chronic muscle inflammation and weakness. Muscle weakness is typically proximal and generally worsens over a subacute or chronic timeframe. Polyarthritis is also often present. Although this patient has weakness, his concomitant paresthesias and ascending pattern is more consistent with GBS.

[» Feedback](#)

79% **C** Schwann cells



Guillain-Barré syndrome is an autoimmune syndrome triggered by an antecedent infection, for example, Campylobacter enteritis (likely in this patient) or CMV infection. The most common subtype, acute inflammatory demyelinating



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5% **B** Muscle endomysium

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> Feedback

79% **C** Schwann cells

Guillain-Barré syndrome is an autoimmune syndrome triggered by an antecedent infection, for example, *Campylobacter* enteritis (likely in this patient) or CMV infection. The most common subtype, acute inflammatory demyelinating polyradiculopathy (AIDP), likely arises from cross-reactivity between epitopes from the antecedent infection and the myelin in Schwann cells (also known as molecular mimicry), leading to immune-mediated destruction of Schwann cells and Schwann cell-produced myelin. Infiltration of leukocytes results in demyelination and subsequent axonal degeneration that is worst at the peripheral nerve roots (radiculopathy) and can affect all nerve types (sensory, motor, sympathetic, parasympathetic).



Guillain-Barré syndrome

> Feedback

6% **D** Presynaptic terminals of cholinergic neurons

Clostridium botulinum toxin inhibits the release of ACh at the presynaptic terminus of cholinergic motor and sensory neurons. This toxin has an affinity for stimulatory neurons and thus results in flaccid, rather than spastic (as seen in tetanus), paralysis. It is transmitted via spore infection (commonly found in honey) or ingestion of the toxin itself. Symptoms include the 4 D's: diplopia, dysphagia, dysarthria, and dyspnea, none of which is present in this patient.

> Feedback

3% **E** Cerebral white matter

Cerebral white matter receives its color because of an abundance of myelin. Multiple sclerosis (MS) leads to a destruction of cerebral white matter and demyelination. Although elevated spinal protein is found in both GBS and MS, patients with MS do not present with ascending bilateral weakness. Rather, MS is characterized by neuritis, hemiparesis, hyperreflexia, and/or hemisensory deficits.

> Feedback

A 45-year-old man is brought to the emergency department because of a 1-day history of malaise and abdominal pain. Six weeks ago, he had vomiting and watery diarrhea for 2 days that resolved without treatment. Twelve weeks ago, he underwent orthotopic liver transplantation for alcoholic cirrhosis. At the time of discharge, his total serum bilirubin concentration was 1.0 mg/dL. He stopped drinking alcohol one year ago. His current medications include daily tacrolimus, prednisone, valganciclovir, and trimethoprim-sulfamethoxazole. His temperature is 37.7°C (99.9°F), pulse is 95/min, and blood pressure is 150/80 mm Hg. He appears uncomfortable and has mild jaundice. Examination shows scleral icterus. The abdomen is soft and tender to deep palpation over the right upper quadrant, where there is a well-healed surgical scar. His leukocyte count is 2500/mm³, serum bilirubin concentration is 2.6 mg/dL, and serum tacrolimus concentration is within therapeutic range. Which of the following is the next appropriate step in diagnosis?

[» Feedback](#)

- ☐ A | Magnetic resonance cholangiopancreatography
- ☐ B | CT scan of the abdomen with contrast
- ☐ C | Viral loads
- ☐ D | Liver biopsy
- ☐ E | Endoscopic retrograde cholangiopancreatography
- ☐ F | Esophagogastroduodenoscopy
- ☐ G | Ultrasound of the liver

[Show answer](#)

%



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[» Feedback](#)

Acute liver transplant rejection can occur within the first 6 months after transplantation and may present with nausea/vomiting, jaundice, and abdominal pain!



5% **A** Magnetic resonance cholangiopancreatography

Magnetic resonance cholangiopancreatography (MRCP) would be appropriate if the first-line diagnostic study suggested complications affecting the biliary tree. MRCP is not the first diagnostic modality in the work up of suspected acute liver transplant rejection, as it visualizes the biliary tract only and is furthermore time-consuming and expensive.

[» Feedback](#)

11% **B** CT scan of the abdomen with contrast

CT scan of the abdomen with contrast is not the first-line radiologic study in the workup of suspected acute liver transplant rejection. Only if the first-line diagnostic modality suggests vascular complications (e.g., hepatic artery thrombosis), biliary complications (e.g., leak, stricture), or liver infection (e.g., perihepatic abscess), a CT scan would be appropriate. However, a more cost-effective method without radiation and contrast exposure should be conducted first.

[» Feedback](#)

12% **C** Viral loads

Viral infection (e.g., HSV, CMV or EBV) can present with jaundice, abdominal pain, and elevated bilirubin, as seen in this patient.



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12% **C** Viral loads

Viral infection (e.g., HSV, CMV or EBV) can present with jaundice, abdominal pain, and elevated bilirubin, as seen in this patient. However, this patient is taking CMV prophylaxis (valganciclovir), making a viral infection less likely. The rather unspecific clinical presentation of this patient (pain in the graft region, jaundice, and rising bilirubin) without associated fever raises concern for acute transplant rejection or other complications of liver transplantation (e.g., portal vein thrombosis, biliary leak, liver infection), which should be evaluated with a different diagnostic tool first. If infection remains the suspected cause of symptoms, testing for viral loads (viral particles detected by PCR) is appropriate.

[» Feedback](#)20% **D** Liver biopsy

Liver biopsy is the gold standard for the definitive diagnosis of acute transplant rejection and will likely be required in this patient at a later time. However, biopsy is an invasive procedure associated with small but considerable risk of complications, especially in liver transplant recipients. It can demonstrate pathologies of the liver parenchyma, but not of the extrahepatic vessels and biliary ducts. Since this patient's clinical findings are rather nonspecific, another less invasive diagnostic tool should be used first to rule out other potential post-transplant complications.

[» Feedback](#)5% **E** Endoscopic retrograde cholangiopancreatography

Endoscopic retrograde cholangiopancreatography provides excellent visualization of the biliary tree and would be appropriate in this patient (i.e., < 6 months post-transplant with hepatic dysfunction) after the first-line diagnostic tool suggested biliary pathology (e.g., dilation, stricture, leak). Since this patient's presentation is rather non-specific, and can be caused by other complications of liver transplantation (acute transplant rejection, vascular complications, infection), a more cost-effective and less invasive diagnostic modality should be performed first.

[» Feedback](#)



12% **C** Viral loads

Viral infection (e.g., HSV, CMV). However, this patient is taking antiviral therapy for a clinical presentation of this concern for acute transplant liver infection), which should be considered. Symptoms, testing for viral



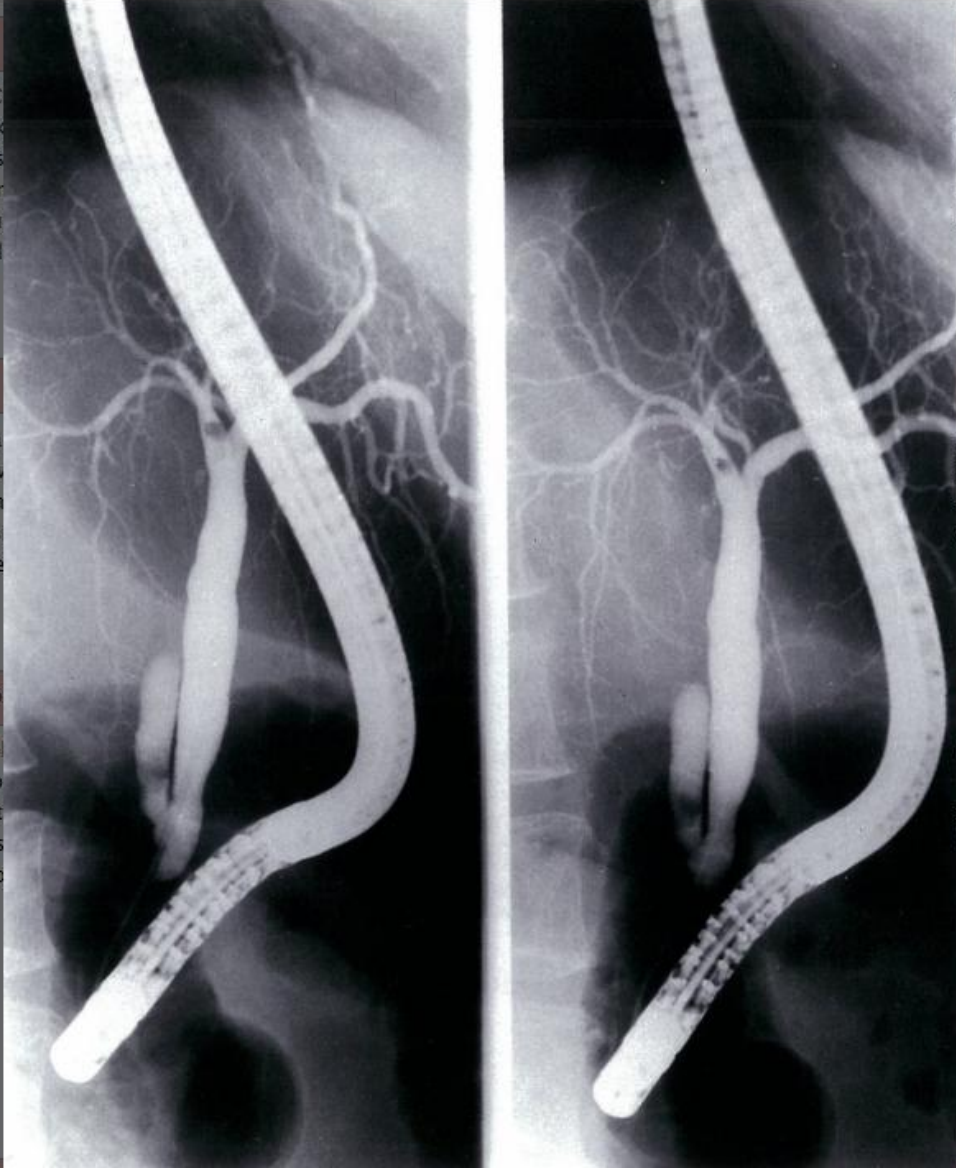
20% **D** Liver biopsy

Liver biopsy is the gold standard for this patient at a later time. However, especially in liver transplant recipients, it is important to rule out biliary complications and biliary ducts. Staining should be used first to rule



5% **E** Endoscopic retrograde cholangiopancreatography

Endoscopic retrograde cholangiopancreatography (ERCP) is the gold standard for this patient (i.e., < 6 months post-transplant). Pathology (e.g., dilation, strictures) of liver transplant complications of liver transplant should be used first to rule out less invasive diagnostic modalities.



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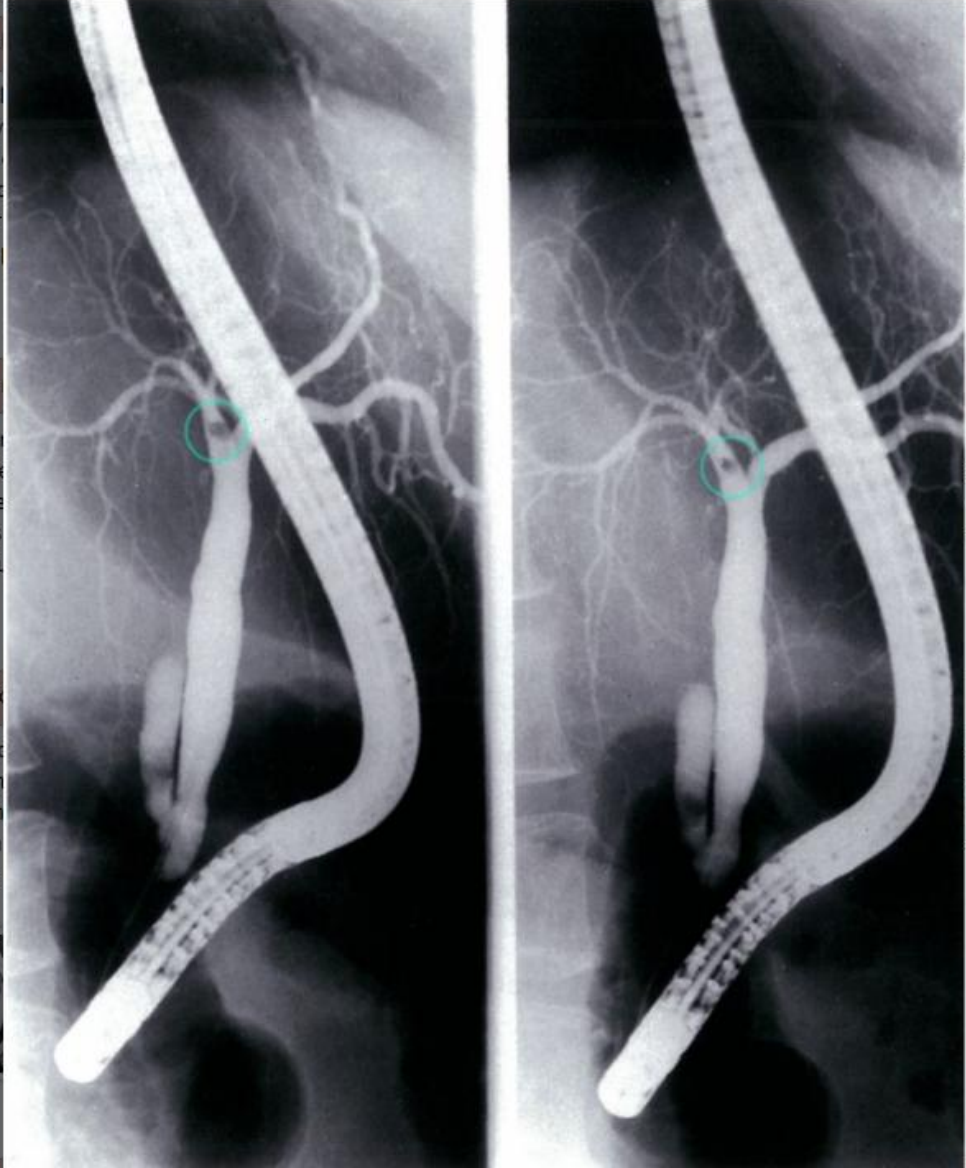
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> Feedback

likely be required in this erable risk of complications, ut not of the extrahepatic invasive diagnostic tool

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ee and would be appropriate nostic tool suggested biliary nd can be caused by other n), a more cost-effective and

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Status post cholecystectomy: ERCP shows concrements in the hepatic duct

Lateral view: a small concrement is located in the right hepatic duct just before it unites with the left hepatic duct to form the common hepatic duct.

Further description:
The spine is seen on the left side of the image; the radiopaque endoscope reaches into the image from above. In the centre one can identify the intrahepatic and extrahepatic bile ducts. A remainder of the cystic duct s/p cholecystectomy can be seen lateral to the common bile duct.

Endoscopic retrograde cholangiopancreatography provides visualization of the biliary tree and would be appropriate in this patient (i.e., < 6 months post-transplant with hepatic dysfunction) after the first-line diagnostic tool suggested biliary pathology (e.g., dilation, stricture, leak). Since this patient's presentation is rather non-specific, and can be caused by other complications of liver transplantation (acute transplant rejection, vascular complications, infection), a more cost-effective and less invasive diagnostic modality should be performed first.



> Feedback

0% **F** Esophagogastroduodenoscopy

Esophagogastroduodenoscopy is usually performed if, e.g., gastritis, gastrointestinal bleeding, or formation (and/or possible bleeding) of esophageal varices is suspected. The rather unspecific clinical presentation of this patient (pain in the graft region, jaundice, and rising bilirubin) within 6 months after liver transplantation raises concern for acute transplant rejection or other complications of liver transplantation (e.g., portal vein thrombosis, biliary leak, liver infection), which should be evaluated with a different diagnostic tool.



> Feedback

48% **G** Ultrasound of the liver

The presence of pain in the graft region (RUQ), jaundice, and rising bilirubin within 6 months of orthotopic liver transplantation raises concern for acute transplant rejection. However, since these clinical findings are nonspecific, duplex ultrasound of the liver should be performed to rule out possible vascular complications (e.g. portal vein thrombosis, bleeding), biliary complications (e.g., leak, stricture), and liver infection (e.g. perihepatic abscess). Duplex ultrasound is a non-invasive, precise, and cost-effective method that can be used at the patient's bedside. If acute transplant rejection is suspected, a liver biopsy showing interstitial lymphocytic infiltrate would confirm the diagnosis.



Transplantation

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A 35-year-old man comes to the physician because of a 6-month history of fatigue and increased sweating at night. He says that he feels "constantly tired" and needs more rest than usual although he sleeps well. In the morning, his sheets are often wet and his skin is clammy. He has not had any sore throat, runny nose, or cough recently. He has not traveled anywhere. Over the past 4 months, he has had a 6.8-kg (15-lb) weight loss, despite having a normal appetite. He does not drink or urinate more than usual. He is 181 cm (5 ft 11 in) tall and weighs 72 kg (159 lb); BMI is 22 kg/m². His temperature is 37.9°C (100.2°F), pulse is 65/min, and blood pressure is 120/70 mm Hg. Physical examination shows no abnormalities. An HIV screening test and confirmatory test are both positive. The CD4 count is 600 cells/μl and the viral load is 10⁴ copies/mL. Treatment with lamivudine, zidovudine, and indinavir is begun. The patient is at greatest risk for which of the following adverse effects?



» Feedback



- ☐ A | Stevens-Johnson syndrome
- ☐ B | Hepatic failure
- ☐ C | Hypersensitivity reaction
- ☐ D | Hyperpigmentation of palms and soles
- ☐ E | Pancreatitis
- ☐ F | Chronic kidney disease
- ☒ G | Urolithiasis



Show answer

%

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[» Feedback](#)

This patient, who has been diagnosed with HIV infection and started on combination antiretroviral therapy, is at greatest risk of developing an adverse effect from indinavir therapy.



9% **A** Stevens-Johnson syndrome

Stevens-Johnson syndrome is an adverse effect that more commonly occurs with non-nucleoside reverse transcriptase inhibitors, such as nevirapine or efavirenz, than other antiretroviral drug groups.

[» Feedback](#)

21% **B** Hepatic failure

Hepatotoxicity is an adverse effect of nevirapine, a non-nucleoside reverse transcriptase inhibitor (NNRTI). Other side effects of NNRTIs include CNS toxicity (associated with efavirenz) and Stevens-Johnson syndrome. This patient, however, is not taking an NNRTI.

[» Feedback](#)

8% **C** Hypersensitivity reaction

Hypersensitivity can occur with any drug but is a particularly worrisome known adverse effect of abacavir, an nucleoside reverse transcriptase inhibitor (NRTI). This potentially life-threatening syndrome of fever and maculopapular rash occurs most commonly in patients who are HLA-B*57:01 positive. Thus, testing for HLA-B*57:01 must be performed prior to starting any



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» Feedback

 3% **D** Hyperpigmentation of palms and soles

Hyperpigmentation is an adverse effect of emtricitabine, a nucleoside reverse transcriptase inhibitor (NRTI), but not of lamivudine, zidovudine, or indinavir.

» Feedback

 28% **E** Pancreatitis

Pancreatitis (both acute and chronic) is a known adverse effect of didanosine, as well as stavudine (didanosine carries a black box warning for pancreatitis). Both drugs are nucleoside reverse transcriptase inhibitors. Pancreatitis is not a known adverse effect of lamivudine, indinavir, or even zidovudine (despite it also being a NRTI).

» Feedback

 8% **F** Chronic kidney disease

Chronic kidney disease is an adverse effect that among antiretroviral drugs most commonly occurs with tenofovir, a nucleotide reverse transcriptase inhibitor. Chronic kidney disease is not a typical adverse effect of lamivudine, zidovudine, or indinavir.

» Feedback

 22% **G** Urolithiasis ✓

Indinavir is a protease inhibitor that has poor solubility and significant renal excretion, which leads to crystallization and stone formation (urolithiasis) in up to 20% of patients.



Human immunodeficiency virus

» Feedback